

TPU0S1T processed

SDOF peak-picking modal fit after Schmitz & Smith, Machining Dynamics, 2nd ed., Sec. 2.5.1 (Fig. 2.21, Eqs. 2.58, 2.60, 2.62, 2.63)

1. Test

File	/home/user/TPU/Transmissibility test/0%/TPU0S1T.csv
Experiment / FRF type	transmissibility (Y axis units in file: m/s^2 / m/s^2)
Conversion applied	none (transmissibility, accel/accel, dimensionless)
Sign convention	+1 (Fig. 2.21: Im < 0 at resonance)
Data	2047 points, 0.391-799.614 Hz
Search band	30-500 Hz
Smoothing	11-point quadratic Savitzky-Golay (real and imaginary parts; raw maximum unsmoothed)
Processed	2026-10-02 21:21

2. Natural frequency and picked points (Fig. 2.21, one mode)

Point	Meaning	Variable	f [Hz]	omega [rad/s]	Value	Coherence
raw max	maximum of the raw FRF magnitude	omega_n_raw	55.078	346.07	FRF = 4.809 [-]	0.752
1	minimum of the imaginary part	omega_n1, A	58.203	365.70	A = -3.036 [-]	0.604
3	biggest maximum of the real part	omega_3	53.907	338.70	Re = 3.828 [-]	0.653
4	biggest minimum of the real part	omega_4	74.219	466.33	Re = -1.124 [-]	0.713

3. Modal parameters

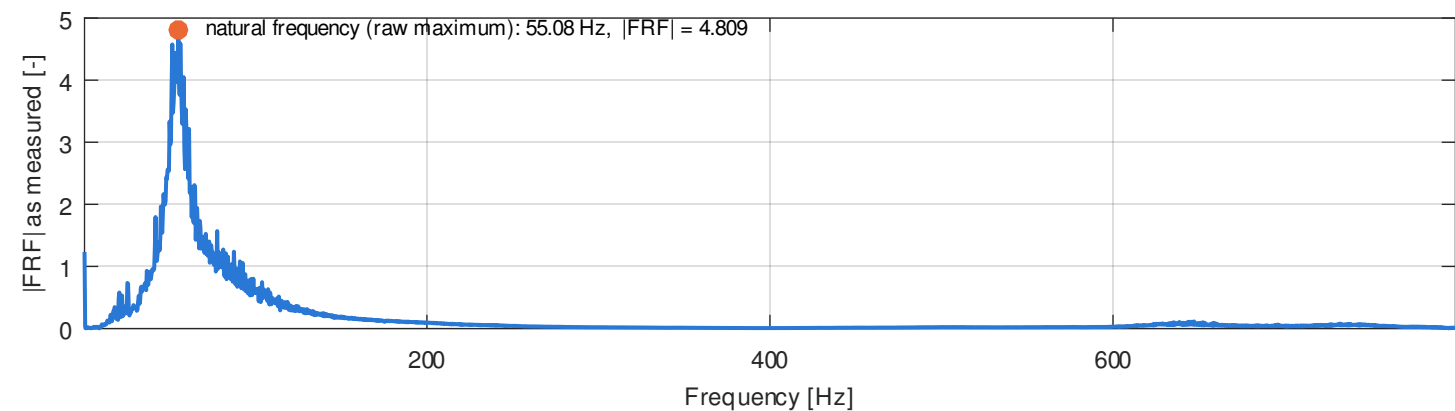
Equation	Quantity	Variable	Value	Units
2.58	damping ratio	$\text{zeta}_{q1} = (\omega_4 - \omega_3) / (2 \omega_{n1})$	0.1745	-
2.60	stiffness	$k_{q1} = -1 / (2 \text{zeta}_{q1} A)$	0.9438	-
2.62	mass	$m_{q1} = k_{q1} / \omega_{n1}^2$	7.057e-06	s^2
2.63	damping	$c_{q1} = 2 \text{zeta}_{q1} \sqrt{k_{q1} m_{q1}}$	0.0009007	s

4. Notes

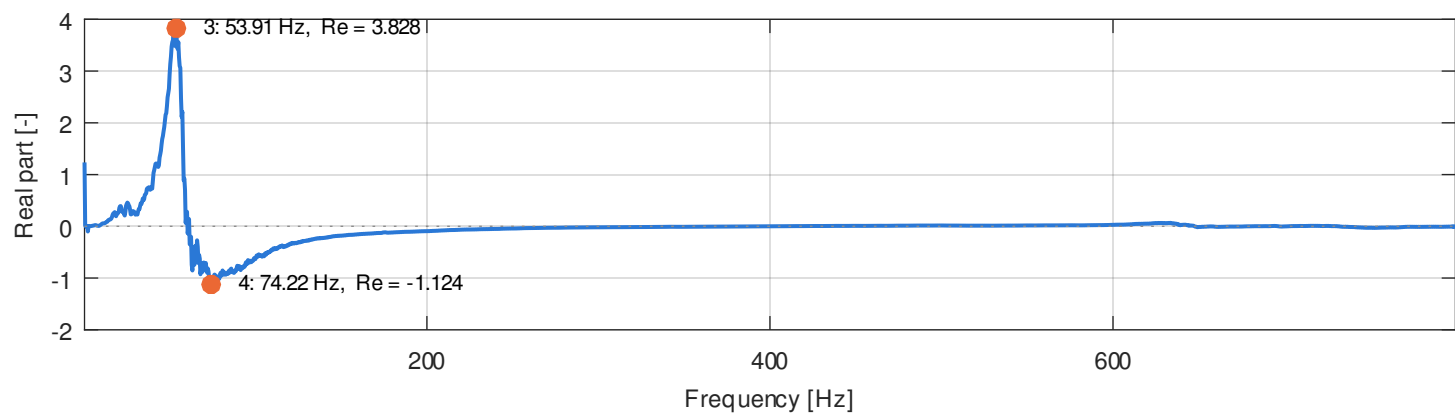
- Transmissibility has no force: A is dimensionless and k_{q1} , m_{q1} , c_{q1} come out in [-], [s^2], [s], not N/m, kg, N s/m. For one mode on a moving base $\text{Im}(T) = -1 / (2 \text{zeta})$ at resonance, so Eq. 2.60 gives k_{q1} close to 1.
- Coherence below 0.75 at 1, 3, 4 (0.60, 0.65, 0.71): those values are uncertain.
- Points 3 and 4 are not symmetric about point 1 (4.30 Hz below, 16.02 Hz above); for one viscous mode they lie near $\omega_{n1} (1 \mp \text{zeta}_{q1})$.

5. Raw FRF magnitude, real and imaginary parts

TPU0S1T: raw FRF magnitude (no smoothing)



TPU0S1T: real part (smoothed: 11-point Savitzky-Golay)



TPU0S1T: imaginary part (smoothed: 11-point Savitzky-Golay)

